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Shading Idea 1: Discrete Tone Mapping with Matrix Distortion

This model utilizes discrete thresholding for tone mapping and applies a color distortion matrix in specific intensity ranges to achieve a stylized look.

$$\begin{aligned}\mathbf{L}_c, \mathbf{L}_v &\in \mathbb{R}^3 \quad (\text{Light Direction, Light Color Vector}) \\ \mathbf{T}(u, v) &: [0, 1]^2 \rightarrow \mathbb{R}^3 \quad (\text{Texture Color Map}) \\ \hat{n} &\in \mathbb{R}^3, \quad \|\hat{n}\| = 1 \quad (\text{Surface Normal Vector}) \\ L &= \max(0, -\mathbf{L}_c \cdot \hat{n}) \quad (\text{Lambertian Intensity})\end{aligned}$$

The tone mapping function $h_f(L)$ is defined as:

$$h_f(L) = \begin{cases} 1 & L > x_1 \\ L & x_1 \geq L > x_2 \\ c_1 & x_2 \geq L > x_3 \\ 0 & x_3 \geq L \end{cases}$$

The color distortion vector function $\mathbf{v}(\mathbf{L}_v)$ is defined as:

$$\mathbf{v}(\mathbf{L}_v) = \begin{cases} \mathbf{L}_v & L > x_1 \\ \mathbf{A}\mathbf{L}_v & x_1 \geq L > x_2 \\ \mathbf{L}_v & x_2 \geq L \end{cases} \quad (\text{Warm Distortion Matrix})$$

The final color is computed using the Hadamard product ($\odot$):

$$\text{Color}_{\text{final}} = h_f(L) \cdot [\mathbf{T}(u, v) \odot \mathbf{v}(\mathbf{L}_v)]$$

Parameters: $x_1, x_2, x_3 \in \mathbb{R}$ (Thresholds), $c_1 \in \mathbb{R}$ (Shadow Intensity), $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ (Color Matrix).

Shading Idea 2.1: Continuous Gradient Matrix Mapping

This model introduces continuous gradient matrices $\mathbf{A}(L)$ and $\mathbf{B}(L)$ at the boundaries of tone transitions to ensure smooth chromatic aberration or warm-tone shifts.

$$\begin{aligned}h_f(L) &= \begin{cases} 1 & L > x_1 \\ c_1 & x_1 \geq L > x_2 \\ 0 & x_2 \geq L \end{cases} \\ \mathbf{v}(\mathbf{L}_v) &= \begin{cases} \mathbf{A}(L)\mathbf{L}_v & \delta_{1,1} \geq L > \delta_{1,2} \quad (\text{Warm/Dark Gradient Region}) \\ \mathbf{B}(L)\mathbf{L}_v & \delta_{2,1} \geq L > \delta_{2,2} \quad (\text{Warm/Bright Gradient Region}) \\ \mathbf{L}_v & \text{otherwise} \end{cases}\end{aligned}$$

Continuity Constraints:

$$\begin{aligned}\mathbf{A}(L), \mathbf{B}(L) &\in C^k([0, 1], \mathbb{R}^{3 \times 3}) \quad \text{where } k \geq 1 \\ \mathbf{A}(\delta_{1,1}) &= \mathbf{A}(\delta_{1,2}) = \mathbf{B}(\delta_{2,1}) = \mathbf{B}(\delta_{2,2}) = \mathbf{I}_{3 \times 3}\end{aligned}$$

Parameter Valid Intervals: To ensure proper gradient transition regions between tones, the threshold parameters must satisfy the following inequality chain:

$$1 \geq \delta_{1,1} > \delta_{1,2} > x_1 > \delta_{2,1} > \delta_{2,2} > x_2 > 0$$

- $[\delta_{1,2}, \delta_{1,1}]$: Gradient region before transitioning to the mid-tone (x_1).
- $[\delta_{2,2}, \delta_{2,1}]$: Gradient region before transitioning to the shadow tone (x_2).

$$\text{Color}_{\text{final}} = h_f(L) \cdot [\mathbf{T}(u, v) \odot \mathbf{v}(\mathbf{L}_v)]$$